

CSE 340: Computer Architecture

Answer the following questions. Show calculations where required. Figures in the right margin are part of the examination.

1.
CO1

A processor executes a program, which has a total of 30×10^9 instructions, categorized as follows:

Instruction Type	Instruction Count	Cycle per instruction
R-type	50% of total instructions	2
I-type	20% of total instructions	4
SB-type	20% of total instructions	3
U-type	10% of total instructions	1

- a) Find the average CPI of the program. 2
 b) If the CPU clock frequency is 3.2 GHz, calculate the execution time of the program. 1

$$CPI = \frac{\text{Inst} \times \text{Avg}}{\text{freq}}$$

2.
CO1

- a) Define throughput and response time in computer systems. What does it mean for a system to have high throughput with high response time? 2
 b) A program runs on CPU M in 100s. CPU N optimizes 60% of the instructions to run 1.5 times faster, while the remaining instructions execute at the same speed. Calculate the new execution time of CPU N. 2

$$(1 - 60) \times 100 = 30$$

3.
CO1

Read each statement carefully. Determine if the following sentences/statements are true or false. For any false sentence/statement, write its correct form.

- a) -0xB,4A3 can be stored precisely in the following format. 1
- | | | |
|--------------|-------------------|--------------------|
| Sign (1 bit) | Exponent (5 bits) | Fraction (10 bits) |
|--------------|-------------------|--------------------|
- b) If a CPU with 2.5x instruction count and 4GHz clock speed performs the same task in 10 seconds as another CPU with 1.5x instruction count and 4.5GHz clock speed, then the CPI of the first CPU must be lower. 1
- c) A 16-core processor can always achieve a 16x speedup over a single-core version on any program. 1

$$2^5 - 1 = 31$$

4.
CO1

$(1.1111011011)_2 \times 2^{17}$. You want to store this floating point number using IEEE754 floating point representation. You have two format options:

	Sign Bit	Exponent	Fraction
Option - 1	1 bit	7 bits	6 bits
Option - 2	1 bit	6 bits	7 bits

- a) If your goal is to achieve precision, which of the two options would you choose? Justify your answer. 2
- b) What is the range of usable actual exponent that can be stored in a 24-bit IEEE format where 6 bits are reserved for the fraction? 2

5.
CO2

- a) Construct the equivalent RISC-V assembly code for the following C program: 5

<pre>arr[18] = func(8, 16); c = arr[19] + d int func(int a, int b) { if (a <= b) { while (a != b) { p += 2; } b = b & 1; } else { p = b << 4; } return (p + b); }</pre>	Variables c and d are stored in registers X₁₈ and X₁₉ respectively. Function arguments a and b are stored in registers X₁₀ and X₁₁ respectively. Variable p in the function func, also uses register X₁₉. arr is a halfword array. The base address of arr is stored in X₂₀. Hint: << means left shift slli
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b)

Instruction Address	Instruction
0xB0A1834A	bne X ₂₁ , X ₂₂ L0.
.....	
.....	
.....	
.....	
0xB0A1838A	L0: add X ₁ , X ₄ , X ₂₀

Convert the instruction, bne X₂₁, X₂₂ L0; to machine code. For bne, opcode = 1100111, funct3 = 001; Your final answer must be in Hexadecimal.

- c) Construct the equivalent RISC-V assembly code for the following C program: 3

<pre>a = 0xD035B1BC b = a / 32</pre>	a and b variables are stored in X₁₉, X₂₀ registers, respectively.
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Handwritten notes:

- 6 bits
- 2x4 = 16
- X₂₂, X₂₂ L0
- 0x B0
- a/32
- 32 2⁵
- slli